

§2.3 恰当方程与积分因子

一、恰当方程的定义及条件

设 $u = u(x, y)$ 是一个连续可微的函数,则它的全微分为

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

如果恰好有方程

$$\frac{\partial u(x, y)}{\partial x} dx + \frac{\partial u(x, y)}{\partial y} dy = 0$$

就可以马上写出它的隐式解

$$u(x, y) = c.$$

1 恰当方程的定义

定义1 若有函数 $u(x, y)$, 使得

$$du(x, y) = M(x, y)dx + N(x, y)dy$$

则称微分方程

$$M(x, y)dx + N(x, y)dy = 0, \quad (1)$$

是恰当方程. 此时(1)的通解为 $u(x, y) = c$.

如 $d(xy) = xdy + ydx = 0$

$$d(x^3y + xy^2) = (3x^2y + y^2)dx + (x^3 + 2xy)dy = 0$$

$$d\left(\int f(x)dx + \int g(y)dy\right) = f(x)dx + g(y)dy = 0$$

是恰当方程.

需考虑的问题

$$M(x, y)dx + N(x, y)dy = 0, \quad (1)$$

- (1) 方程(1)是否为恰当方程?
- (2) 若(1)是恰当方程,怎样求解?
- (3) 若(1)不是恰当方程,有无可能转化为恰当方程求解?

2 方程为恰当方程的充要条件

定理1 设函数 $M(x, y)$ 和 $N(x, y)$ 在一个矩形区域 R 中连续且有连续的一阶偏导数,则方程

$$M(x, y)dx + N(x, y)dy = 0, \quad (1)$$

为恰当方程的充要条件是

$$\frac{\partial M(x, y)}{\partial y} = \frac{\partial N(x, y)}{\partial x}, \quad (2).$$

证明 “必要性” 设(1)是恰当方程, 则有函数 $u(x, y)$, 使得

$$du(x, y) = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy = M(x, y)dx + N(x, y)dy$$

故有 $\frac{\partial u}{\partial x} = M(x, y), \quad \frac{\partial u}{\partial y} = N(x, y)$

从而 $\frac{\partial M}{\partial y} = \frac{\partial^2 u}{\partial y \partial x}, \quad \frac{\partial N}{\partial x} = \frac{\partial^2 u}{\partial x \partial y}.$

由于 $\frac{\partial^2 u}{\partial y \partial x}$ 和 $\frac{\partial^2 u}{\partial x \partial y}$ 都是连续的, 从而有 $\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 u}{\partial x \partial y},$

故 $\frac{\partial M(x, y)}{\partial y} = \frac{\partial N(x, y)}{\partial x}.$

“充分性” 若 $\frac{\partial M(x, y)}{\partial y} = \frac{\partial N(x, y)}{\partial x}$,

则需构造函数 $u(x, y)$, 满足

$$du(x, y) = M(x, y)dx + N(x, y)dy, \quad (4)$$

即应满足 $\frac{\partial u}{\partial x} = M(x, y), \quad (5)$

$$\frac{\partial u}{\partial y} = N(x, y), \quad (6)$$

从(5)出发, 把 y 看作参数, 解这个方程得

$$u(x, y) = \int M(x, y)dx + \varphi(y).$$

$$u(x, y) = \int M(x, y)dx + \varphi(y).$$

$$\frac{\partial u}{\partial y} = N(x, y), \quad (6)$$

这里 $\varphi(y)$ 是 y 的任意可微函数,

下面选择 $\varphi(y)$,使 u 同时满足(6),即

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y} \int M(x, y)dx + \frac{d\varphi(y)}{dy} = N$$

因此

$$\frac{d\varphi(y)}{dy} = N - \frac{\partial}{\partial y} \int M(x, y)dx \quad (7)$$

下面证明(7)的右端与 x 无关, 即对 x 的偏导数常等于零

事实上

$$\begin{aligned} \frac{\partial}{\partial x} \left[N - \frac{\partial}{\partial y} \int M(x, y)dx \right] \\ = \frac{\partial N}{\partial x} - \frac{\partial}{\partial x} \left[\frac{\partial}{\partial y} \int M(x, y)dx \right] \end{aligned}$$

$$= \frac{\partial N}{\partial x} - \frac{\partial}{\partial y} \left[\frac{\partial}{\partial x} \int M(x, y) dx \right] = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \equiv 0.$$

于是, (7)右端的确只含有 y , 积分之得

$$\varphi(y) = \int \left[N - \frac{\partial}{\partial y} \int M(x, y) dx \right] dy,$$

故

$$\frac{d\varphi(y)}{dy} = N - \frac{\partial}{\partial y} \int M(x, y) dx \quad (7)$$

$$u(x, y) = \int M(x, y) dx + \int \left[N - \frac{\partial}{\partial y} \int M(x, y) dx \right] dy, \quad (8)$$

即 $u(x, y)$ 存在, 从而(1)为恰当方程。

注: 若(1)为恰当方程, 则其通解为

$$\int M(x, y) dx + \int \left[N - \frac{\partial}{\partial y} \int M(x, y) dx \right] dy = c, \quad c \text{ 为任常数}$$

二、恰当方程的求解：关键是求原函数的问题。

1 不定积分法

1⁰ 判断 $M(x, y)dx + N(x, y)dy = 0$ 是否为恰当方程,
若是进入下一步.

2⁰ 求 $u(x, y) = \int M(x, y)dx + \varphi(y)$,

3⁰ 由 $\frac{\partial u}{\partial y} = N(x, y)$ 求 $\varphi(y)$.

例1 验证方程

$$(e^x + y)dx + (x - 2 \sin y)dy = 0$$

是恰当方程,并求它的通解.

解: 这里 $M(x, y) = e^x + y$, $N(x, y) = x - 2 \sin y$.

$$\text{所以 } \frac{\partial M(x, y)}{\partial y} = 1 = \frac{\partial N(x, y)}{\partial x},$$

故所给方程是恰当方程.

由于所求函数 $u(x, y)$ 满足

$$\frac{\partial u}{\partial x} = e^x + y, \quad \frac{\partial u}{\partial y} = x - 2 \sin y,$$

由偏导数的定义, 只要将 y 看作常数, 将 $e^x + y$ 对 x 积分得

$$u(x, y) = \int (e^x + y) dx + \varphi(y) = e^x + yx + \varphi(y).$$

$$u(x, y) = e^x + yx + \varphi(y).$$

对 $u(x, y)$ 关于 y 求偏导数,得 $\varphi(y)$ 应满足的方程为

$$x + \frac{d\varphi(y)}{dy} = x - 2\sin y$$

$$\text{即 } \frac{d\varphi(y)}{dy} = -2\sin y$$

积分后得: $\varphi(y) = 2\cos y,$

$$\text{故 } u(x, y) = e^x + yx + 2\cos y.$$

从而方程的通解为

$$e^x + yx + 2\cos y = c.$$

2 线积分法

定理1充分性的证明也可用如下方法:

$$\text{由于 } \frac{\partial M(x, y)}{\partial y} = \frac{\partial N(x, y)}{\partial x},$$

由数学分析曲线积分与路径无关的定理知:

$M(x, y)dx + N(x, y)dy$ 为某函数 $u(x, y)$ 的全微分,

即有函数 $u(x, y)$,使

$$du(x, y) = M(x, y)dx + N(x, y)dy,$$

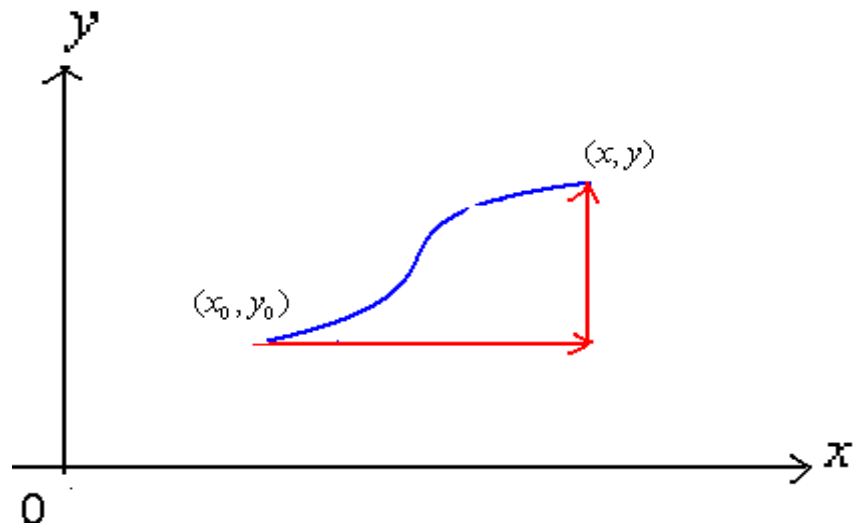
从而(1)为恰当方程。

Green 公式

设 $D \subset \mathbb{R}^2$ 为有界区域, 边界 ∂D 为分段光滑的简单闭曲线, 取正向 (沿曲线行进时区域在左侧)。若 $P, Q \in C^1(\overline{D})$, 则

$$\oint_{\partial D} (P dx + Q dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

这时, 取 $(x_0, y_0) \in R$, 则



$$\begin{aligned} u(x, y) &= \int_{(x_0, y_0)}^{(x, y)} M(x, y) dx + N(x, y) dy \\ &= \int_{x_0}^x M(x, y_0) dx + \int_{y_0}^y N(x, y) dy, \end{aligned}$$

从而(1)的通解为

$$\int_{x_0}^x M(x, y_0) dx + \int_{y_0}^y N(x, y) dy = c, \quad c \text{ 为任意常数。}$$

例2 求解方程

$$(y \cos x + 2xe^y)dx + (\sin x + x^2e^y + 2)dy = 0.$$

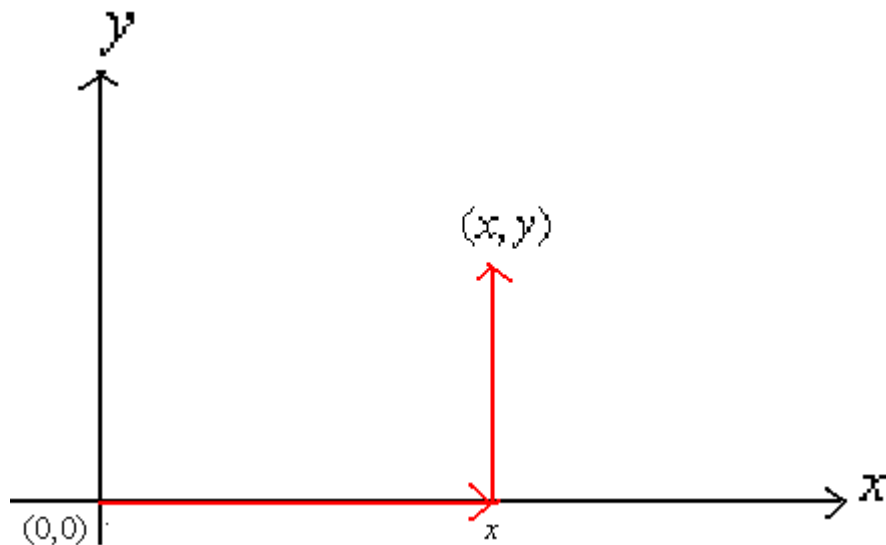
解: 由于 $M(x, y) = y \cos x + 2xe^y$, $N(x, y) = \sin x + x^2e^y + 2$,

$$\frac{\partial M(x, y)}{\partial y} = \cos x + 2xe^y = \frac{\partial N(x, y)}{\partial x},$$

故所给方程是恰当方程.

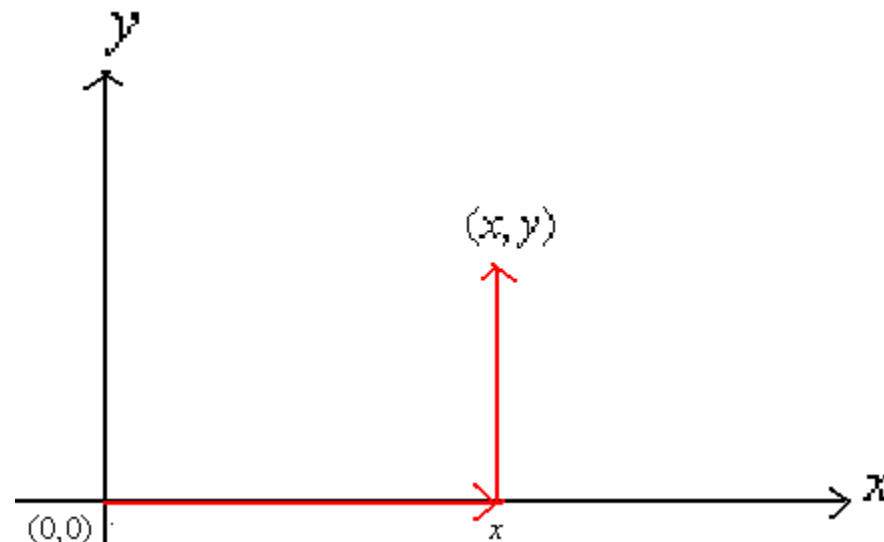
由于 $M(x, y), N(x, y)$ 在全平面上连续,

故取 $(x_0, y_0) = (0, 0)$, 则



$$M(x, y) = y \cos x + 2xe^y$$

$$N(x, y) = \sin x + x^2 e^y + 2,$$



$$\begin{aligned}
 u(x, y) &= \int_{(0,0)}^{(x,y)} M(x, y)dx + N(x, y)dy \\
 &= \int_0^x M(x, 0)dx + \int_0^y N(x, y)dy \\
 &= \int_0^x 2xdx + \int_0^y (\sin x + x^2 e^y + 2)dy \\
 &= x^2 + y \sin x + x^2 (e^y - 1) + 2y. \\
 &= y \sin x + x^2 e^y + 2y.
 \end{aligned}$$

故通解为: $y \sin x + x^2 e^y + 2y = c$, c 为任常数.

3分组凑微法

采用“**分项组合**”的方法,把本身已构成全微分的项分出来,再把剩余的项凑成全微分.

---应熟记一些简单二元函数的全微分.

如

$$ydx + xdy = d(xy),$$

$$\frac{ydx - xdy}{y^2} = d\left(\frac{x}{y}\right),$$

$$\frac{-ydx + xdy}{x^2} = d\left(\frac{y}{x}\right),$$

$$\frac{ydx - xdy}{xy} = d\left(\ln \left| \frac{x}{y} \right| \right),$$

$$\frac{ydx - xdy}{x^2 + y^2} = d\left(\arctan \frac{x}{y}\right),$$

$$\frac{ydx - xdy}{x^2 - y^2} = \frac{1}{2} d\left(\ln \left| \frac{x - y}{x + y} \right| \right).$$

例3 求方程 $(3x^2 + 6xy^2)dx + (6x^2y + 4y^3)dy = 0$ 的通解.

解: 这里 $M(x, y) = 3x^2 + 6xy^2$, $N(x, y) = 6x^2y + 4y^3$,

$$\text{所以 } \frac{\partial M(x, y)}{\partial y} = 12xy = \frac{\partial N(x, y)}{\partial x},$$

故所给方程是恰当方程. 把方程重新“分项组合”得

$$3x^2dx + 4y^3dy + (6xy^2dx + 6x^2ydy) = 0$$

$$\text{即 } dx^3 + dy^4 + (3y^2dx^2 + 3x^2dy^2) = 0$$

$$\text{或写成 } d(x^3 + y^4 + 3x^2y^2) = 0$$

故通解为: $x^3 + y^4 + 3x^2y^2 = c$, c 为任常数。

例4 验证方程

$$(\cos x \sin x - xy^2)dx + y(1 - x^2)dy = 0,$$

是恰当方程,并求它满足初始条件 $y(0)=2$ 的解.

解: 这里 $M(x, y) = \cos x \sin x - xy^2$, $N(x, y) = y(1 - x^2)$,

$$\frac{\partial M(x, y)}{\partial y} = -2xy = \frac{\partial N(x, y)}{\partial x},$$

故所给方程是恰当方程. 把方程重新“**分项组合**”得

$$\cos x \sin x dx - (xy^2 dx + x^2 y dy) + y dy = 0,$$

$$\text{即 } d\frac{1}{2}\sin^2 x - d\frac{1}{2}x^2 y^2 + d\frac{1}{2}y^2 = 0,$$

$$d \frac{1}{2} \sin^2 x - d \frac{1}{2} x^2 y^2 + d \frac{1}{2} y^2 = 0$$

或写成 $d(\sin^2 x - x^2 y^2 + y^2) = 0$,

故通解为: $\sin^2 x - x^2 y^2 + y^2 = c$,

由初始条件 $y(0) = 2$, 得 $c = 4$,

故所求的初值问题的解为:

$$\sin^2 x - x^2 y^2 + y^2 = 4.$$

三、积分因子

非恰当方程如何求解？

对变量分离方程：

$$dy - f(x)\varphi(y)dx = 0, \quad \text{不是恰当方程.}$$

方程两边同乘以 $\frac{1}{\varphi(y)}$, 得

$$\frac{1}{\varphi(y)} dy - f(x)dx = 0,$$

$$\frac{\partial(-f(x))}{\partial y} = 0 = \frac{\partial \frac{1}{\varphi(y)}}{\partial x}$$

是恰当方程.

对一阶线性方程:

$$dy - (P(x)y + Q(x))dx = 0, \quad \text{不是恰当方程.}$$

方程两边同乘以 $e^{-\int P(x)dx}$, 得

$$e^{-\int P(x)dx} dy - e^{-\int P(x)dx} (P(x)y + Q(x))dx = 0,$$

则
$$\frac{\partial e^{-\int P(x)dx}}{\partial x} = -p(x)e^{-\int P(x)dx} = \frac{\partial -e^{-\int P(x)dx} (p(x)y + Q(x))}{\partial y}$$

$$\text{或左边} = d(e^{-\int P(x)dx} y - \int Q(x)e^{-\int P(x)dx} dx) = 0,$$

是恰当方程.

可见,对一些非恰当方程,乘上一个因子后,可变为恰当方程.

$$M(x, y)dx + N(x, y)dy = 0, \quad (1)$$

1 定义 如果存在连续可微函数 $\mu(x, y) \neq 0$, 使得

$$\mu(x, y)M(x, y)dx + \mu(x, y)N(x, y)dy = 0$$

为恰当方程, 则 $\mu(x, y)$ 是方程(1)的一个积分因子.

例5 验证 $\mu(x, y) = x^2 y$ 是方程

$$(3y + 4xy^2)dx + (2x + 3x^2 y)dy = 0$$

的一个积分因子, 并求其通解.

解: 对方程有

$$\mu(x, y)M(x, y) = 3x^2 y^2 + 4x^3 y^3$$

$$\mu(x, y)N(x, y) = 2x^3 y + 3x^4 y^2$$

由于

$$\frac{\partial \mu(x, y) M(x, y)}{\partial y} = 6x^2 y + 12x^2 y^2 = \frac{\partial \mu(x, y) N(x, y)}{\partial x}$$

故所给方程乘于 $\mu(x, y)$ 后为恰当方程,

所以 $\mu(x, y)$ 是其积分因子.

对方程两边同乘以 $\mu(x, y) = x^2 y$ 后得

$$(3x^2 y^2 + 4x^3 y^3)dx + (2x^3 y + 3x^4 y^2)dy = 0$$

把以上方程重新“分项组合”得

$$(3x^2 y^2 dx + 2x^3 y dy) + (4x^3 y^3 dx + 3x^4 y^2 dy) = 0$$

$$\text{即} \quad dx^3 y^2 + dx^4 y^3 = 0$$

也即 $d(x^3 y^2 + x^4 y^3) = 0$

故所给方程的通解为:

$$x^3 y^2 + x^4 y^3 = c, \quad c \text{ 为任意常数。}$$

2 积分因子的确定

$\mu(x, y)$ 是方程 $M(x, y)dx + N(x, y) = 0$ 的积分因子的充要条件是:

$$\frac{\partial \mu(x, y) M(x, y)}{\partial y} = \frac{\partial \mu(x, y) N(x, y)}{\partial x}$$

即
$$N \frac{\partial \mu}{\partial x} - M \frac{\partial \mu}{\partial y} = \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \mu$$

$$N \frac{\partial \mu}{\partial x} - M \frac{\partial \mu}{\partial y} = \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \mu$$

上面方程是以 $\mu(x, y)$ 为未知函数的偏微分方程, 要想从以上方程求出 $\mu(x, y)$, 一般来说比直接解微分方程 $M(x, y)dx + N(x, y)dy = 0$ 更困难.

尽管如此, 方程

$$N \frac{\partial \mu}{\partial x} - M \frac{\partial \mu}{\partial y} = \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \mu$$

还是提供了寻找特殊形式积分因子的途径.

如果方程 $M(x, y)dx + N(x, y) = 0$ 存在仅与 x 有关的积分因子 $\mu(x, y) = \mu(x)$, 则 $\frac{\partial \mu}{\partial y} = 0$, 这时方程

$$N \frac{\partial \mu}{\partial x} - M \frac{\partial \mu}{\partial y} = \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \mu$$

变成

$$N \frac{d\mu}{dx} = \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \mu$$

即

$$\frac{d\mu}{\mu} = \frac{\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right)}{N} dx$$

由于上式左侧仅与 x 有关,
所以上式右侧只能是 x 的函数的微分,

从而微分方程 $M(x, y)dx + N(x, y) = 0$ 有一个仅依赖于 x 的积分因子的必要条件是

$$\frac{\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right)}{N}, \quad (10)$$

$$\frac{d\mu}{\mu} = \psi(x)dx$$

若(10)只是 x 的函数 $\psi(x)$, 而与 y 无关.

$$\text{则 } \mu(x) = e^{\int \psi(x) dx}, \quad \text{这里 } \psi(x) = \frac{\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right)}{N}$$

是方程 $M(x, y)dx + N(x, y)dy = 0$ 的一个积分因子。

$$\begin{aligned}
\frac{\partial \mu(x)N(x, y)}{\partial x} &= N(x, y) \frac{d\mu(x)}{dx} + \mu(x) \frac{\partial N(x, y)}{\partial x} \\
&= N(x, y) e^{\int \Psi(x) dx} \Psi(x) + \mu(x) \frac{\partial N(x, y)}{\partial x} \\
&= \left(\frac{\partial M(x, y)}{\partial y} - \frac{\partial N(x, y)}{\partial x} \right) \mu(x) + \mu(x) \frac{\partial N(x, y)}{\partial x} \\
&= \mu(x) \frac{\partial M(x, y)}{\partial y} = \frac{\partial \mu(x)M(x, y)}{\partial y}
\end{aligned}$$

故 $\mu(x)$ 是方程 $M(x, y)dx + N(x, y)dy = 0$ 一个积分因子.

3 定理 微分方程

$$M(x, y)dx + N(x, y)dy = 0, (1)$$

有一个仅依赖于x的积分因子的充要条件是

$$\frac{\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right)}{N},$$

仅与x有关,这时(1)的积分因子为

$$\mu(x) = e^{\int \psi(x) dx}, \quad \text{这里 } \psi(x) = \frac{\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right)}{N}$$

同理,微分方程(1)有一个仅依赖于 y 的积分因子的充要条件是

$$\frac{\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right)}{-M},$$

仅与 y 有关,这时(1)的积分因子为

$$\mu(y) = e^{\int \varphi(y) dy}, \quad \text{这里 } \varphi(y) = \frac{\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right)}{-M}.$$

例6 求微分方程 $(\frac{y^2}{2} + 2ye^x)dx + (y + e^x)dy = 0$ 的通解.

解: 这里 $M(x, y) = \frac{y^2}{2} + 2ye^x$, $N(x, y) = y + e^x$,

$$\text{由于 } \frac{\partial M(x, y)}{\partial y} = y + 2e^x \neq \frac{\partial N(x, y)}{\partial x} = e^x,$$

故它不是恰当方程,

$$\text{又由于 } \frac{(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x})}{N} = \frac{y + e^x}{y + e^x} = 1 = \psi(x)$$

它与 y 无关,故方程有一个仅与 x 有关的积分因子

$$\mu(x) = e^{\int \psi(x) dx} = e^{\int 1 dx} = e^x$$

$$\psi(x) = \frac{\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right)}{N} = 1$$

对方程两边同乘以 $\mu(x) = e^x$ 后得

$$\left(\frac{y^2}{2}e^x + 2ye^{2x}\right)dx + (ye^x + e^{2x})dy = 0$$

利用恰当方程求解法得通解为

$$\frac{y^2}{2}e^x + ye^{2x} = c, \quad c \text{ 为任意常数.}$$

- 积分因子是求解积分方程的一个极为重要的方法
- 绝大多数方程求解都可以通过寻找到一个合适的积分因子来解决
- 但求微分方程的积分因子十分困难,需要灵活运用各种微分法的技巧和经验

例7 求解方程

$$\frac{dy}{dx} = -\frac{x}{y} + \sqrt{1 + \left(\frac{x}{y}\right)^2}, \quad (y > 0).$$

解: 方程改写为: $xdx + ydy = \sqrt{x^2 + y^2} dx$,

$$\text{或: } \frac{1}{2} d(x^2 + y^2) = \sqrt{x^2 + y^2} dx,$$

易看出,此方程有积分因子 $\mu(x, y) = \frac{1}{\sqrt{x^2 + y^2}}$,

以 $\mu(x, y)$ 乘改写后的方程两边得:

$$\frac{d(x^2 + y^2)}{2\sqrt{x^2 + y^2}} = dx,$$

即 $d\sqrt{x^2 + y^2} = dx,$

$$\frac{d(x^2 + y^2)}{2\sqrt{x^2 + y^2}} = dx,$$

故方程的通解为: $\sqrt{x^2 + y^2} = x + c,$ c 为任常数.

例8 求解方程

$$ydx + (y - x)dy = 0.$$

解: 这里 $M(x, y) = y, N(x, y) = y - x,$

$$\frac{\partial M(x, y)}{\partial y} = 1 \neq \frac{\partial N(x, y)}{\partial x} = -1,$$

故方程不是恰当方程,

方法1: 因为 $\frac{(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x})}{-M} = -\frac{2}{y} = \varphi(y)$ 仅与 y 有关,

故方程有一个仅依赖于 y 的积分因子

$$\mu(y) = e^{\int \varphi(y) dy} = e^{\int -\frac{2}{y} dy} = \frac{1}{y^2},$$

以 $\mu = \frac{1}{y^2}$ 乘方程两边得: $\frac{1}{y} dx + \frac{1}{y} dy - \frac{x}{y^2} dy = 0.$

$$\text{即 } \frac{ydx - xdy}{y^2} + \frac{dy}{y} = 0.$$

故方程的通解为: $\frac{x}{y} + \ln|y| = c.$

方法2: 方程改写为: $ydx - xdy = -ydy$,

容易看出方程左侧有积分因子:

$$\mu = \frac{1}{y^2} \text{ 或 } \frac{1}{x^2} \text{ 或 } \frac{1}{xy} \text{ 或 } \frac{1}{x^2 + y^2} \dots$$

但方程右侧仅与 y 有关,

故取 $\mu = \frac{1}{y^2}$ 为方程的积分因子,由此得

$$\frac{ydx - xdy}{y^2} = -\frac{dy}{y}.$$

故方程的通解为: $\frac{x}{y} + \ln|y| = c.$

方法3: 方程改写为:

$$\frac{dy}{dx} = \frac{y}{x-y} = \frac{\frac{y}{x}}{1-\frac{y}{x}}$$

这是齐次方程, 令 $u = \frac{y}{x}$ 代入方程得

$$x \frac{du}{dx} + u = \frac{u}{1-u}, \quad \text{即} \quad \frac{1-u}{u^2} du = \frac{1}{x} dx,$$

$$\text{故通解为:} \quad -\frac{1}{u} - \ln|u| = \ln|x| + c,$$

$$\text{变量还原得原方程的通解为:} \quad \frac{x}{y} + \ln|y| = c.$$

方法4: 方程改写为:

$$\frac{dx}{dy} = \frac{1}{y}x - 1,$$

它是以 x 为未知函数, y 为自变量的一阶线性微分方程

$$\begin{aligned}\text{故方程的通解为: } x &= e^{\int p(y)dy} \left(\int Q(y)e^{-\int p(y)dy} dy + \tilde{c} \right) \\ &= e^{\int \frac{1}{y} dy} \left(-\int e^{-\int \frac{1}{y} dy} dy + \tilde{c} \right) \\ &= y \left(-\int \frac{1}{y} dy + \tilde{c} \right) = y(-\ln|y| + \tilde{c}),\end{aligned}$$

即方程的通解为: $\frac{x}{y} + \ln|y| = c.$

作业

• P41

1: (1) , (5)

2: (1) , (4) , (7) , (9)

3, 4, 5, 6, 9